OpenMobile: Building Open Mobile Agents with Task and Trajectory Synthesis

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Paper version: arXiv 2604.15093v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Leading mobile agents keep their training data and synthesis recipes closed.*

Open-data agents sit at around **30%** on AndroidWorld, far behind closed-data systems, and the field cannot study why.

Motivation: three gaps

1. **Closed recipes.** Data and synthesis are undisclosed.
2. **Coupled tasks.** One trajectory bounds the instruction.
3. **No recovery data.** Expert traces show no mistakes.

Evaluation setup

AndroidWorld · 116 | AndroidLab · 138 | MobileWorld · 201 | Qwen2.5-VL-7B | Qwen3-VL-8B | Gemini-3.1-Pro expert | 3 runs | SFT, 3 epochs

- **All online** benchmarks; no static sets.
- **Not fully held out:** data made in the AndroidWorld emulator.

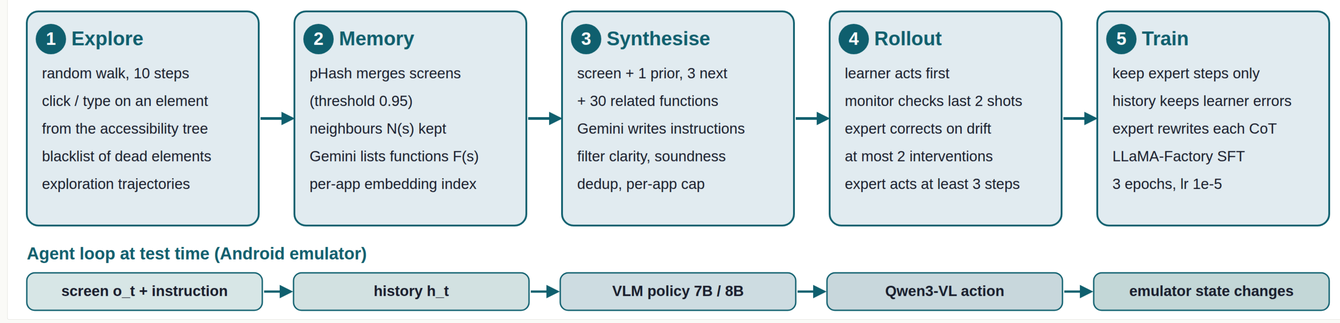
Our read (CUA Radar)

73	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; 3 runs; open baselines; ablations; code. **Partly:** held-out apps, budget, cost.
- **Flags:** AP5, data made in the AndroidWorld environment. **Not reported:** a check of the LLM filter.
- Pattern K15. Authors' numbers, not rerun.

Method: explore, build a screen memory, synthesise tasks, switch policies

- **Sees / acts:** screen, instruction and history in; Qwen3-VL action space and response format out.
- **Tasks:** instructions composed from a global screen memory, not one trajectory.
- **Signal:** expert steps only; learner errors stay in the history as context.



Our redraw (not the paper's figure): the five stages from random-walk exploration to SFT, and the agent loop.

$$a_t \sim \pi_{z_t}(\cdot \mid I, o_t, h_t), \quad z_t \in \{e, l\}$$

z_t picks the expert e or the learner l at each step

Key results: online Pass@1 (%)

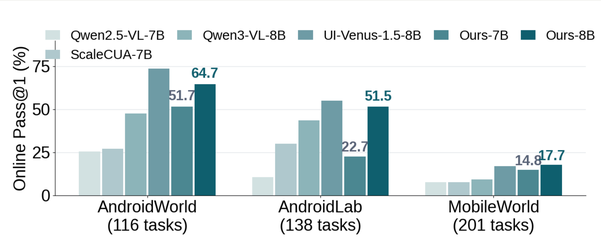


Table 1, redrawn: Pass@1, mean of 3 runs; MobileWorld GUI-only subset.

AndroidWorld: Ours-8B 64.7 against 47.6 for its Qwen3-VL-8B base model

SO WHAT → *Pass@1 only; half-ranges and Pass@3 are in the paper's Table 1.*

Ablation: rollout strategy, AndroidWorld Pass@1 (%)

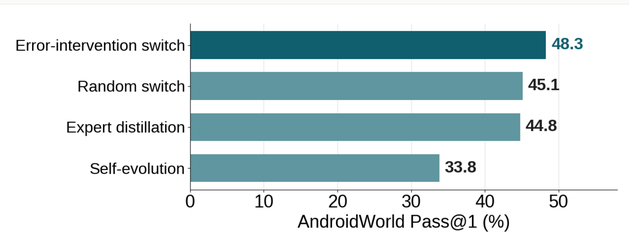


Table 3, redrawn: Qwen2.5-VL-7B learner, 1.5K trajectories per strategy.

SO WHAT → *Error-intervention switching is best, with the most recovery steps per trajectory (1.56 vs 0.42 for expert distillation; hand-counted on 50). Self-evolution is weakest (33.8). **Task synthesis**, same 1.5K budget: OpenMobile 48.3, coupled 45.3, OS-Genesis 34.1. All at the 7B learner, AndroidWorld.*

Headline Numbers

64.7%
AndroidWorld Pass@1
Qwen3-VL-8B base: 47.6%

- **51.7%** 7B, AndroidWorld
- **51.5%** 8B, AndroidLab
- **17.7%** 8B, MobileWorld

The paper's own controls

CONTROL (ONLINE)	OPENMOBILE	COMPARED WITH
Synthesis, 1.5K traj.	48.3	Coupled 45.3; OSG 34.1
AndroidLab, 7B	22.7	ScaleCUA-7B 30.0
RL on 8B	RL 64.1	SFT 64.7
Similarity > 0.7	3.5% of tasks	AndroidControl, AMEX

7B trails ScaleCUA-7B on AndroidLab

Takeaway

Open synthesis lifts both bases on three online benchmarks; the 7B model trails ScaleCUA-7B on AndroidLab, and data is made inside the AndroidWorld environment.

2.8K tasks, 34K steps | 20 Android apps | 12.2 steps per trajectory | data and code released

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